RL-Based Gain Optimization of Time-Delay Control for AUVs with a Pan-Tilt Sensor Pointing Mechanism

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Abstract : Autonomous underwater vehicles (AUVs) equipped with a two-axis pan-tilt mechanism (PTM) for sensor pointing are subjected to reaction forces and moments generated by PTM motion, which can degrade vehicle pose stabilization and sensor-pointing performance. This paper uses reinforcement learning (RL) to optimize the six feedback gains of a Time-Delay Control (TDC) baseline and the intensity of an active vibration damper, training the policy with a staged curriculum in simulation. Compared with the best grid-searched fixed gains, the learned policy reduces position RMSE by 20.5% and vibration energy by 35.7%, with a slight improvement in attitude RMSE. It also outperforms an oracle rule-based switcher. Analysis of the learned gains shows, however, that the policy converges to a near-constant operating point rather than scheduling gains according to the disturbance regime. Therefore, the performance improvement results from identifying a better operating point rather than from online adaptation.

1. Introduction

Autonomous underwater vehicles (AUVs) are increasingly equipped with actively steered sensor pointing mechanisms, such as camera or sonar mounts, that retarget independently of the vehicle body (Fig. 1). Every retargeting motion produces an inertial reaction wrench on the body that falls in the same low-frequency band as the attitude and position loops, degrading pose-holding accuracy and sensor-pointing stability. Because the reaction is strong while the two-axis pan-tilt mechanism (PTM) scans and nearly absent while it is parked, a single set of fixed gains is necessarily a compromise between the two regimes.

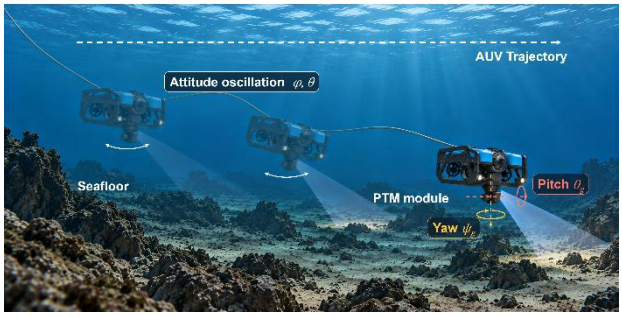


Fig. 1 Concept of an AUV equipped with a two-axis pan-tilt sensor pointing mechanism. Its rotation generates reaction forces and moments that disturb the vehicle's position and attitude.

Time-Delay Control (TDC) [1] estimates unmodeled dynamics and disturbances from one-step-delayed input and acceleration measurements and cancels them in the control law, and it already absorbs a large part of this structured reaction wrench. Learning an additional compensation wrench on top of TDC therefore leaves little headroom. Reinforcement learning (RL) has instead been used to tune controller gains, e.g., a PPO-trained [3] policy that adapts the cascaded gains of a quadcopter in flight [4]. Following this direction, this paper uses RL to optimize the TDC feedback gains and the intensity of an active vibration damper, and then analyzes what the learned policy actually does.

The contributions of this paper are: (1) an RL-based optimization of six TDC gains and one damper intensity for a PTM-equipped AUV, trained with a staged curriculum; (2) a comparison with the best grid-searched fixed gains and an oracle rule-based switcher, in which the learned policy reduces position RMSE by 20.5% and vibration energy by 35.7% relative to the best fixed baseline; and (3) an analysis of the learned gain trajectory showing that the improvement comes from a near-constant operating point rather than state-dependent gain scheduling.

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2. System Modeling and Baseline Control

2.1. AUV–PTM Coupled Dynamics

The vehicle carries a two-axis PTM for sensor pointing (Fig. 2). The coupled system follows the standard 6-DOF equation of motion [2]

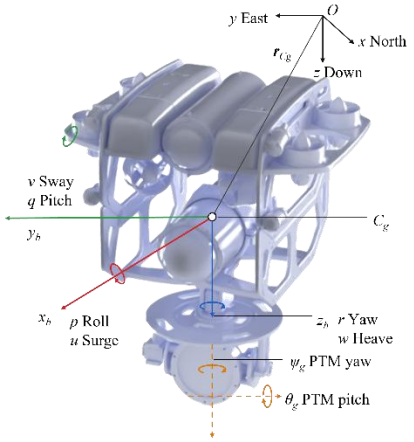


Fig. 2 BlueROV2 Heavy configuration with the two-axis PTM: NED frame, body frame (x_b, y_b, z_b), and PTM pan angle ψ_g and tilt angle θ_g.

$$M(\eta)\dot{\nu} + C(\nu)\nu + D(\nu)\nu + g(\eta) = \tau + \tau_g, \qquad (1)$$

where η and ν are the vehicle pose and body-frame velocity, M, C, D, and g are the mass (including added mass), Coriolis, damping, and restoring terms, τ is the control wrench, and τ_g is the PTM reaction wrench. From conservation of the PTM's angular momentum,

$$\tau_g \approx A\ddot{q}_g + C_d \left|\dot{q}_g\right| \dot{q}_g, \qquad (2)$$

where A and C_d depend on the retargeting profile. Because these coefficients are hard to identify online, τ_g is not modeled in the controller; it is treated as part of the lumped disturbance estimated by TDC.

2.2. Time-Delay Control and Vibration Damping

With p, R, v_b, and ω the position, attitude, body linear velocity, and body angular rate, and subscript d denoting desired values, the tracking errors are

$$e_p = p_{\mathrm{d}} - p, \quad e_v = v_{\mathrm{d}} - R v_{\mathrm{b}}, \tag{3}$$

$$e_R = \frac{1}{2}\left(R_{\mathrm{d}}^{\mathrm{T}} R - R^{\mathrm{T}} R_{\mathrm{d}}\right)^{\vee}, \quad e_\omega = \omega_{\mathrm{d}} - \omega, \tag{4}$$

where e_R is the SO(3) geometric attitude error [7]. Desired accelerations follow a PD-plus-feedforward law,

$$\alpha_v = R^{\mathrm{T}}\left(a_{\mathrm{d}} + K_{\mathrm{d,pos}} e_v + K_{\mathrm{p,pos}} e_p\right), \tag{5}$$

$$\alpha_\omega = K_{\mathrm{d,att}} e_\omega - K_{\mathrm{p,att}} e_R, \quad \nu_{\mathrm{r}} = [\alpha_v; \alpha_\omega]. \tag{6}$$

The disturbance is estimated from the previous step's acceleration and delivered wrench and injected into the control law with a diagonal trust gain Λ,

$$\hat{H}(t) \approx H(t-L) = \hat{M}\dot{v}(t-L) - \tau(t-L), \tag{7}$$

$$\tau = \hat{M}\nu_{\mathrm{r}} - \Lambda\hat{H}(t), \quad \Lambda = \mathrm{diag}\left(\lambda_{\mathrm{lin}} 1_3, \lambda_{\mathrm{ang}} 1_3\right), \tag{8}$$

where $\hat{M}$ is a nominal inertia estimate and L is the time delay. The four feedback gains and λ_{lin}, λ_{ang} are the six controller parameters optimized in Sec. 3. The wrench τ is distributed to the eight thrusters by a damped pseudo-inverse with smooth saturation, and the saturated wrench is fed back to the time-delay estimate to prevent windup.

A separate damper feeds the band-pass-filtered body rate (f_0 = 3.0 Hz, Q = 0.7) back as an attitude-only torque,

$$\omega_{\mathrm{BP}}(t) = \mathrm{BP}_{f_0,Q}\left(\omega(t)\right), \quad \tau_{\mathrm{damper}} = [0; -\alpha_t K_0 \omega_{\mathrm{BP}}(t)], \tag{9}$$

where $\alpha_t \in [0, 1.5]$ is the damper intensity set by the policy.

3. RL-Based Gain Optimization

3.1. Policy Architecture and Action Space

A policy π_θ, trained with PPO [3], sets the controller parameters at every control step from the vehicle and PTM state (Fig. 3). The TDC law and the damper of Sec. 2.2 are unchanged; the policy acts only through their parameters.

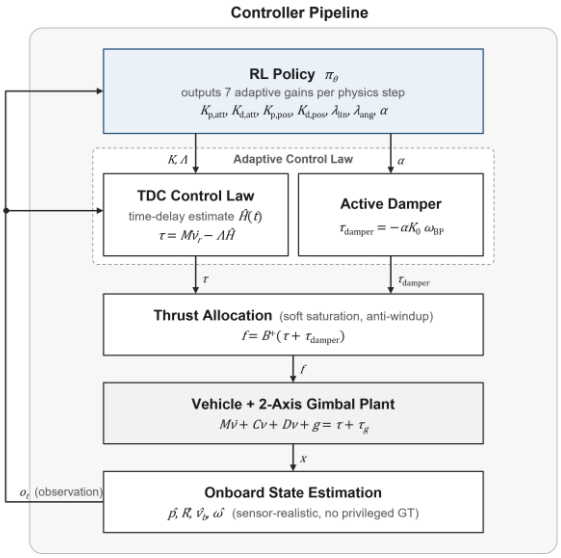


Fig. 3 Controller design: the RL policy sets the TDC gains and damper intensity from the onboard-estimated state, and thrust allocation and the AUV–PTM plant close the loop.

The action is seven-dimensional: $K_{p,att}$, $K_{d,att}$, $K_{p,pos}$, $K_{d,pos}$, λ_{lin}, λ_{ang}, and α. Each raw output is squashed to $z_t \in [0, 1]$ by a sigmoid and mapped to a validated safe range (Table 1),

$$g_t = g_{\mathrm{lo}} + z_t\left(g_{\mathrm{hi}} - g_{\mathrm{lo}}\right), \tag{10}$$

and the damper intensity is scaled as $\alpha_t = \alpha_{max}\, z_{\alpha,t}$. A rate limit on z_t prevents abrupt gain changes.

3.2. Observation and Reward Design

The 68-dimensional observation consists of (i) the tracking errors e_p, e_v, e_R and body rate ω, (ii) the PTM joint rate, acceleration, and pointing error, (iii) a vibration-energy statistic, (iv) the normalized TDC disturbance estimate and its change, (v) thruster-utilization statistics, and (vi) the previous action. All terms are taken from the onboard state estimator rather than simulator ground truth. The per-step cost penalizes tracking error and vibration,

$$\mathrm{J}_t = w_{att}\left\|\mathrm{e}_R\right\|^2 + w_{pos}\left\|\mathrm{e}_p\right\|^2 + w_{vib}\left\|\omega_{BP}\right\|^2 + J_{reg}, \tag{11}$$

where J_{reg} collects small penalties on damper effort, net energy injected by the damper, thruster utilization above a safety margin, and gain-change magnitude. The reward is $r_t = -J_t$, with an additional penalty c_{safety} when an episode is terminated for safety reasons (thruster overload, excessive tracking error, or numerical failure). The weights are listed in Table 2.

4. Simulation and Experimental Results

4.1. Experimental Setup and Baselines

Training and evaluation use MarineGym [6], built on NVIDIA Isaac Sim [5], with a BlueROV2 Heavy (eight thrusters) carrying the two-axis PTM. The control step is Δt = 0.016 s, and training uses 32 parallel environments for 48M frames in total. Tables 1 and 2 list the main parameters.

Parameter	Value / range
Mass $\hat{M}$	20.2 kg
Inertia $\hat{M}$	[1.280, 1.307, 0.504] kg·m^2
$K_{p,att}$ / $K_{d,att}$	[4.0, 10.0] / [1.7, 3.0]
$K_{p,pos}$ / $K_{d,pos}$	[1.5, 3.5] / [2.0, 3.0]
λ_{lin} / λ_{ang}	[0.06, 0.20] / [0.16, 0.40]
α (damper)	[0, 1.5]
L, Δt	0.1 s, 0.016 s

Table 1 Controller parameters and action ranges.

Item	Value
Algorithm, network	PPO, MLP [256, 256, 256]
Learning rate, clip, γ	5×10^{-4}, 0.1, 0.99
Envs, episode length	32, 12000 steps
Curriculum frames	8M + 8M + 8M + 24M
w_{att}, w_{pos}, w_{vib}	1.0, 1.0, 0.1
w_{res}, w_P, w_{act}, w_Δ	0.02, 0.05, 1.0, 0.5
c_{safety}	50

Table 2 Training settings and reward weights.

The baselines were selected by an exhaustive grid search over $K_{p,att}$, K_{pos}, and λ_{ang} for each PTM regime. The best fixed gains are A_{opt} ($K_{p,att}$ = 6.0) for the parked regime and B_{opt} ($K_{p,att}$ = 9.7) for the active regime. The two differ almost only in the attitude gain, so any single fixed gain set is a compromise. The learned policy is compared with Fixed A_{opt}, Fixed B_{opt}, and a rule-based switcher that toggles between A_{opt} and B_{opt} using the true PTM-active flag. The metrics are position RMSE, attitude RMSE, and vibration energy (band-pass power of ω_{BP}), averaged over three matched seeds.

4.2. Control Performance Comparison

Condition	Position RMSE (m)	Attitude RMSE (deg)	Vib. energy
Fixed A_{opt}	0.315	13.09	7.25
Fixed B_{opt}	0.303	10.97	8.83
Rule-based switcher	0.352	11.03	8.81
Learned (ours)	**0.241**	**10.86**	**5.68**

Table 3 Control performance of the baselines and the learned policy (3-seed mean).

As Table 3 shows, the learned policy outperforms every baseline on all three metrics. Relative to B_{opt}, the best fixed baseline, it reduces position RMSE by 20.5% (0.303 → 0.241 m) and vibration energy by 35.7% (8.83 → 5.68), while attitude RMSE improves slightly (10.97° → 10.86°). The oracle switcher does not outperform fixed B_{opt} (position RMSE 0.352 vs. 0.303 m), and a re-sweep of the grid found no fixed gain better than B_{opt} under the mixed schedule.

Fig. 4 shows representative time histories of the tracking errors under the mixed parked/active schedule. Fixed A_{opt}, Fixed B_{opt}, and the rule-based switcher all show large position-error excursions of up to about 0.6 m while the PTM retargets, whereas the learned policy keeps the position error near 0.1 m for most of the run, with only a few short peaks. The attitude error of the learned policy stays close to that of the switcher, which is consistent with the small attitude-RMSE difference in Table 3.

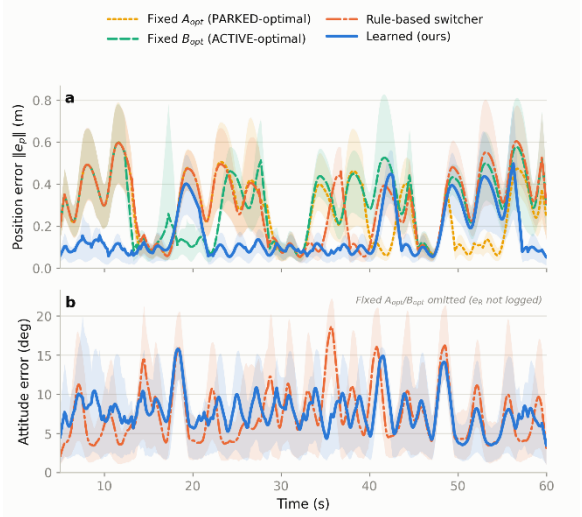


Fig. 4 Time histories of (a) position error $\|e_p\|$ and (b) attitude error for the baselines and the learned policy; shaded bands indicate the variation across seeds. Fixed A_{opt} and B_{opt} are omitted in (b) because e_R was not logged for these runs.

4.3. Analysis of Learned Gain Behavior

The learned gain trajectory shows that the policy does not perform state-dependent scheduling. All seven parameters differ by less than 0.3% between the parked and active regimes (e.g., $K_{p,att}$ = 9.636 vs. 9.644). Vehicle turn rate is strongly correlated with actuator-saturation risk (ρ = 0.82–0.83), yet its correlation with the policy's $K_{p,att}$ output is only ρ ≈ 0.02–0.04. The apparent adaptation early in each episode is an artifact of the rate limit: every episode ramps from the same reset value to the same converged value at the same rate, regardless of the trajectory, to the fourth decimal place across seeds. A weak but consistent effect remains at the extremes of the turn-rate distribution (top vs. bottom decile: +2.7% in $K_{p,att}$, −5.1% in $K_{p,pos}$, −8.1% in λ_{lin}), but it amounts to less than 8% of each parameter's range.

The performance gain therefore comes from a single operating point that dominates every fixed gain found by grid search, not from online adaptation to the disturbance regime. In this setting, RL acted as an effective optimizer of the controller parameters rather than as a gain scheduler.

5. Conclusion

This paper applied RL to optimize the gains and damper intensity of a TDC controller for an AUV subject to PTM reaction disturbances. In simulation, the learned parameters reduced position RMSE by 20.5% and vibration energy by 35.7% relative to the best grid-searched fixed gains, with a slight improvement in attitude RMSE. Analysis of the learned policy, however, showed that it converges to a near-constant operating point rather than scheduling gains according to the state. Achieving genuine state-dependent scheduling, for example by exploiting vehicle turn rate, and validating the controller on the physical BlueROV2 Heavy are left for future work.

Acknowledgment

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References

[1] K. Youcef-Toumi and O. Ito, “A time delay controller for systems with unknown dynamics,” ASME Journal of Dynamic Systems, Measurement, and Control, vol. 112, no. 1, pp. 133-142, 1990.

[2] T. I. Fossen, Handbook of Marine Craft Hydrodynamics and Motion Control, Wiley, 2011.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” arXiv preprint arXiv:1707.06347, 2017.

[4] M. Timmerman, A. Patel, and T. Reinhart, “Adaptive gain scheduling using reinforcement learning for quadcopter control,” arXiv preprint arXiv:2403.07216, 2024.

[5] V. Makoviychuk et al., “Isaac Gym: High performance GPU-based physics simulation for robot learning,” arXiv preprint arXiv:2108.10470, 2021.

[6] S. Chu et al., “MarineGym: A high-performance reinforcement learning platform for underwater robotics,” arXiv preprint arXiv:2503.09203, 2025.

[7] T. Lee, M. Leok, and N. H. McClamroch, “Geometric tracking control of a quadrotor UAV on SE(3),” in Proc. 49th IEEE Conference on Decision and Control (CDC), 2010, pp. 5420-5425.